

Introduction to Neonatology and History Taking

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Introduction to Neonatology / History Taking in Neonatology

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Outline

Definitions · Mortality rates · Causes of neonatal mortality · Preventable factors · History taking
· Examination of the newborn (systems) · Primitive reflexes · APGAR · Resuscitation · Routine
& further care

Definitions

- **Neonatal period** – the **first 28 days of life**
- **Perinatal period** – from the **20th-28th week of gestation to the 1st 7 days of life**

It is a **fragile period** because the infant's **transition from intrauterine to extrauterine life** requires many biochemical and physiologic changes.

Mortality rates (UNICEF 2020)

- **IMR (infant mortality rate)** – 72/1000 live births
- **PMR (perinatal mortality rate)** – 41/1000
- **NMR (neonatal mortality rate)** – 35/1000

Factors contributing to increased PMR: **prenatal, maternal, fetal.**

Causes of neonatal mortality

- Circumstances surrounding delivery
- **IUGR**
- Conditions predisposing to **asphyxia** (placental insufficiency, severe congenital malformation)
- Overwhelming **early-onset neonatal infection**
- **Prematurity and low birth weight**
- **Congenital malformations**

By category:

- **Fetal** – placental insufficiency, intrauterine infection, congenital malformation, cord accidents, abruptio placentae, hydrops fetalis
- **Preterm** – severe immaturity, RDS, IVH, congenital abnormalities, infection, NEC, BPD
- **Full term** – congenital abnormalities, birth asphyxia, trauma, infection, meconium aspiration pneumonia, PPHN

Reducing perinatal/neonatal mortality

Improved obstetric care; reduce preterm birth and LBW; prenatal diagnosis; early management of congenital malformations; timely treatment of diseases from gestation/labour/delivery.

Preventable variables: health education, prenatal care, nutrition, social support, risk identification, obstetric care.

History taking in the newborn

- **Socio/demographic:** name, age, sex; PC; HPC (past medical history is often embodied in the HPC)
- **Pregnancy/birth history:** when pregnancy was noticed, desired or not, any attempt at termination, antenatal events (bleeding, drug use, acute illness), carried to term, how labour started (spontaneous/induced/augmented), prolonged?, membrane rupture time, mode of delivery (normal vaginal or assisted – forceps/vacuum)
- **Immediate neonatal period:** did the baby cry well or need resuscitation; jaundice, cyanosis, pallor, fever, seizures
- **Nutritional:** food type, quantity, quality, frequency
- **Immunisation:** per schedule; sight the card
- **Family:** birth rank and ages of siblings; parental age/occupation/education; family illness (jaundice, diabetes, cardiopulmonary, genetic); previous maternal reproductive problems (stillbirth, preterm, rhesus isoimmunisation, blood group incompatibility)

Examination of the term newborn

Aim: early detection of abnormalities and a **baseline** for subsequent care.

Normal values: weight **2.5-3.5 kg** (~2.9-3 kg in developing countries); length **45-55 cm**; **OFC 35 ± 2 cm**; posture **slightly flexed with clenched fists**.

- **Skin** – vasomotor instability (mottling, acrocyanosis), **lanugo** (more in preterm), **vernix caseosa**, **milia**, **Mongolian blue spots (slate-grey naevi)** over buttocks/back; peeling/cracking in post-term
- **Head** – **anterior fontanelle 0.5-4 cm**, soft, normotensive; craniotabes; **caput succedaneum**, **cephalhaematoma**

- **Face** – asymmetry from in-utero positioning, subconjunctival haemorrhage; responds to sound (startle/blink/cry); check palate and frenulum
- **Neck** – appears short
- **Chest** – **diaphragmatic breathing, RR 30-60/min**, sinus arrhythmia, transient murmur, **BP 45-60 systolic**, palpable breast tissue
- **Abdomen** – flat, weak wall (diastasis recti, umbilical hernias common in black infants); if **scaphoid, rule out diaphragmatic hernia**; liver up to 2 cm palpable
- **Anus** – if **no meconium after 48 hours**, confirm patency (rectal tube/X-ray)
- **Genitalia** – males: transient hydrocoele, prepuce adherent to glans; females: labia majora cover most of labia minora
- **Extremities** – fetal posture effects; movement detects fracture/nerve injury; polydactyly, syndactyly, simian crease; **examine hips for congenital dislocation**
- **Neurologic** – alertness, tone; reflexes – **rooting, sucking, Moro, grasp (palmar and plantar), tonic neck**

APGAR scoring

Sign	0	1	2
Appearance (colour)	Blue	Pink body, blue extremities	Completely pink
Pulse (HR)	Absent	< 100	> 100
Grimace (response)	Nil	Grimace	Cry/resist
Activity (tone)	Limp	Some flexion	Active movement
Respiration (effort)	Absent	Slow/irregular	Strong/good cry

Interpretation: 0-3 severe · 4-6 moderate · 7 and above vigorous.

Resuscitation in the delivery room

- **Routine:** cleaning stimulates the baby (never shake, slap, or turn upside down); cover with fresh linen to keep warm, remove wet linen; if not breathing easily in room air, give oxygen (cupped hands/catheter); monitor HR/RR
- **Not breathing but HR > 60:** ventilate with **bag & mask + O** (or bag & ET tube); proper suctioning (especially meconium risk); ventilation sequence “**BREATHE 2,3, BREATHE 2,3...**” aiming ≥ 60 breaths/min
- **HR < 60: ventilate + adrenaline + external cardiac massage;** sequence “**and 1, and 2, and 3, and BREATHE...**”; monitor every 15 seconds; may need sodium bicarbonate, normal saline, and further NICU care

Further care of the newborn

- **Eyes** – silver nitrate (Credé’s procedure)
- **Cord care** – ligate firmly; **chlorhexidine gel/solution** applied after ligation, daily until the stump drops off
- **Vitamin K** 0.5 mg IM stat
- **Feeding** – commence within **30 minutes** by putting to the breast (encourages let-down); **prelacteal feeds (water/glucose) are wrong practices**

References

Lecture material of Dr Ijeoma Stella; standard neonatology texts.